

Embedded World Offers Plethora of Jobs But With Low Starting Pay

The article discusses the tremendous career opportunities available in the embedded systems industry. It is based mostly on the advice and comments received from experts in the field

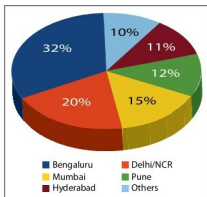
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Embedded systems are popular in healthcare, consumer goods, video security/surveillance, information technology (IT), telecommunication, Internet of Things (IoT), automotive and manufacturing industries, and now in the upcoming wearable electronics sector. With such pronounced existence, let us find out what a career in this field has to offer.

"Embedded design and development industry is so vast that it will consistently need core talented engineers with good hardware and software knowledge and, most importantly, out-of-the-box problem-solving skills," according to Amit Gohel, founder, Eleics Design.

Most electronics graduates are unaware of the abundant opportunities available in the field of embedded industry, believes Sakthivel K., chief executive officer, Caliber Embedded Technologies India (P) Ltd. He says, "According to the NASSCOM report, the total embedded system global business is expected to reach around \$90 billion by 2015, wherein India alone is expected to have a share of around \$16 billion."

On a similar note, Girija VEDI, senior marketing executive, VVDN Technologies also believes that career in the embedded domain is a worthy option available for electronics engineers today. She says, "As VVDN specialises in developing embedded systems for various segments, we are likely to hire around 500 engineers by 2014 year end, but shortage of candidates with the right talent is definitely one of the hurdles we are facing in achieving our goal."



Top five locations hiring professionals for embedded systems (Data courtesy: TimesJobs.com)

Hardware design or software testing option

Fresh electronics engineers having good knowledge and interest in digital and analogue electronics circuit design could choose board design as their field, advises Vinod D. Halappa, senior manager - business development, Dexcel Electronics Designs Pvt Ltd. He says, "With additional knowledge of VLSI/FPGA technology, design flow and HDL/System C, they can also get into ASIC/FPGA design/verification field." Fresh computer engineers with good basics of computer architecture, microprocessor/controller-based system design, embedded C, Linux (or any embedded OS) can go for embedded software as their field. He adds, "There are ample jobs in embedded software testing as well, for which they would need some specific training in state-of-the-art testing tools, methodologies and scripting languages for about three to six months."

With the recent initiatives of the government of India to promote local electronics system design and manufacturing (ESDM), announcements to

setup new semiconductor fabrications units and incentives to units being set up in various electronics clusters, local electronics manufacturing in India is poised for a major growth. Rajeev Tiwari, director, Tevatron Technologies says, "With this renewed emphasis on ESDM, engineers with skills in embedded systems, device drivers, firmware, hardware-software integration, PCB design, board design, graphics driver and real-time operating system would be most sought-after. At the same time, at product level, since a lot of innovation will start taking place in local market, product marketing engineers and managers will also get lot of attention."

With the rapid growth of electronics and consumer markets, plenty of smart devices are getting released within a short span of time. Undoubtedly, there is a shining future ahead for fresh candidates as well, informs Paramita Kapat, head - human resource, eInfochips. She says, "Compact embedded devices are playing a major role in the boom of various industries, especially smartphones, tablets and medical equipments. In nutshell, embedded systems field is the most diversified for electronics as well as computer science students."

Also, with the advancements in electronics, embedded systems field is becoming a niche segment and there is an influx of new products in market. Upendra Patel, CEO, Glide Technology Pvt Ltd says, "This is driving demand for both outsourcing projects as well as domestic products. Due to this, there is good demand for engineers in embedded systems industry." Although, he thinks, there is shortage of qualified and experienced resources.